CAPSTONE PROJECT PHASE-1 (PRESENTATION-2)

Adaptive Human-Review Allocation for Reliable Machine Learning Under Distribution Shift

Course: UQ25CA741A | Department of Computer Applications | PES University, Bangalore

CANDIDATE DETAILS

Candidate SRN: PES1PG25CA101
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Abstract & Core Research Formulation

EXECUTIVE RESEARCH ABSTRACT

• **Operational Bottleneck:** Human-review capacity is strictly constrained in production systems (modeled at 1%, 2%, 5%, and 10% budgets).

• **Silent Failure Mode:** ML models make confident errors when evaluated under real-world temporal distribution shifts [4], [5].

• **Proposed Mechanism:** An adaptive multi-factor allocation framework combining prediction confidence, Mahalanobis OOD distance, PSI feature drift, and historical mistake patterns.

• **Experimental Scope:** Evaluated on a 500,000-flow chronological sample from the BigFlow-NIDS benchmark [11] (66.9M flows, 55 attributes).

• **Empirical Observation:** Multi-factor allocation captures up to 34.26% of errors at 2% budget (vs 26.73% for confidence); confidence alone captures more at 5–10% budgets. The advantage is budget-dependent.

CORE RESEARCH QUESTION

"Can an adaptive, multi-factor human-review allocation mechanism reduce high-risk ML errors more efficiently than fixed-threshold and single-signal strategies under distribution shift and constrained human-review capacity?"

Key Objectives: (1) Synthesize 4 complementary reliability signals. (2) Benchmark under 1%–10% budgets. (3) Evaluate under temporal shift. (4) Deliver functional research software.

Problem Scenario & Proposed Solution

PROBLEM SCENARIO

• **The Attention Bottleneck:**
High-throughput network systems generate millions of predictions daily, but human audit teams can review only 1%–5% due to high operational cost.

• **Silent Overconfidence Failure:**
Under temporal distribution shift, models maintain average confidence over 96.8% despite a 4.55x surge in prediction error rate [4], [5].

• **Inefficacy of Random Audits:**
Uniform random inspection captured only 1.17% of shifted-test errors at a 1% review budget in our experiments.

• **Single-Signal Failure:**
Confidence alone misses novel attack vectors; OOD detection alone flags novelty that does not correspond to classification errors.

PROPOSED SOLUTION

• **Multi-Factor Reliability Engine:**
Synthesizes four complementary signals: prediction uncertainty, Mahalanobis OOD distance, feature drift (PSI), and historical mistake proximity.

• **Budget-Constrained Priority Queue:**
Constructs a composite risk score and allocates top-K predictions matching exact capacity thresholds (1%, 2%, 5%, 10%).

• **Adaptive Signal Weighting:**
Dynamically scales signal weights based on population-level feature drift severity and reviewed error feedback [2], [6].

• **Chronological Shift Evaluation:**
Evaluates the pipeline on temporal partitions to assess performance under genuine temporal degradation, avoiding random split leakage.

Project Purpose & Implementation Scope

PURPOSE OF THE RESEARCH

• **Empirical Investigation:** Determine whether combining heterogeneous reliability signals catches more critical errors than conventional confidence ranking under tight operational budgets.

• **Tackle Overconfidence:** Address the silent failure mode of ML models that misclassify traffic with high confidence under temporal shift [4], [5].

• **Operational Triage Framework:** Provide a principled, reproducible framework for triage routing that explicitly models real-world human inspection constraints (1%, 2%, 5%, 10%).

PROJECT SCOPE (PHASE-1 vs PHASE-2)

• **In-Scope (Phase-1 Implemented):**
Binary classification (Benign vs Attack) on network flow data; 500,000-flow chronological sample from BigFlow-NIDS [11]; simulated review oracle using test ground truth; evaluation across 4 fixed budgets (1%, 2%, 5%, 10%).

• **Out-of-Scope (Phase-1):**
Live production network tap / inline packet processing; actual human reviewers in the loop (simulated ground-truth oracle used); multi-class attack attribution (detailed Attack column excluded to avoid label leakage).

• **Planned Scope (Phase-2):**
Dataset-agnostic tabular generalization across healthcare and financial domains; post-hoc calibration; SHAP explainability for flagged review cases; interactive web dashboard.

Background Study: Existing Systems & Approaches

EXISTING PARADIGMS IN SELECTIVE ML & REVIEW ALLOCATION

• **Confidence-Based Selective Classification [4], [5]:**
Ranks instances strictly by maximum prediction probability max(P(y|x)). Established by Geifman & El-Yaniv (2017) [4]. Limitation: Modern models suffer from severe miscalibration under shift (Guo et al., 2017) [5], making high-confidence errors invisible to this strategy.

• **Out-of-Distribution (OOD) Detection [1], [8]:**
Measures distance from the training distribution in feature space using Mahalanobis or energy metrics (Yang et al., 2024 [8]; Cao et al., 2023 [1]). Limitation: Novel instances do not necessarily correspond to classification errors; OOD alone underperforms random audit in our experiments.

• **Population Distribution Drift Monitoring [1], [9]:**
Detects macroscopic feature shifts across batches using metrics like Population Stability Index (PSI) (Koh et al., 2021 [9]). Limitation: PSI is a population-level aggregate and cannot directly score individual predictions for triage without instance-level projection.

• **Learning to Defer (L2D) [2], [3], [7]:**
Jointly trains a classifier and a rejection policy to defer difficult instances to a human expert (Mozannar & Sontag, 2020 [2]; Wei et al., 2024 [3]; Liu et al., 2022 [7]). Limitation: Most L2D formulations assume fixed deferral costs and stationary data, without explicit fixed-budget operational slicing.

• **Human-in-the-Loop Intrusion Detection [6], [10]:**
Integrates analyst feedback into active learning loops to triage security alerts (Kim et al., IEEE TIFS 2024 [6]; He et al., IEEE COMST 2023 [10]). Motivation: Demonstrates the urgent need for cost-aware routing to alleviate alert fatigue in security operations centers.

Literature Survey: Research Papers (Part 1: Papers [1]–[5])

[1] Cao, Zhu, Pang (2023): **"Anomaly Detection Under Distribution Shift"** **[IEEE & Scopus Indexed]** DOI: 10.1109/ICCV51070.2023.00599
IEEE/CVF International Conference on Computer Vision (ICCV 2023), pp. 6488–6500
Summary: Addresses anomaly detection under shift; directly informs our shifted-test evaluation methodology.

[2] Mozannar & Sontag (2020): **"Consistent Estimators for Learning to Defer to an Expert"** **[Top ML Venue (PMLR)*]** Verified IEEE alternative: C. Geng et al., IEEE TPAMI, 2021
International Conference on Machine Learning (ICML 2020), PMLR vol. 119, pp. 7076–7087
Summary: Theoretical framework for learning-to-defer. Establishes mathematical foundations for human-AI review routing.

[3] Wei, Cao, Feng (2024): **"Exploiting Human-AI Dependence for Learning to Defer"** **[Top ML Venue (PMLR)*]** Verified IEEE alternative: D.-B. Wang et al., IEEE/CVF CVPR, 2022
International Conference on Machine Learning (ICML 2024), PMLR vol. 235, pp. 52484–52499
Summary: Models human-machine interaction in deferral, informing adaptive human allocation strategies.

[4] Geifman & El-Yaniv (2017): **"Selective Classification for Deep Neural Networks"** **[Scopus Indexed Series]** ISSN: 1049-5258
Advances in Neural Information Processing Systems (NeurIPS 2017), vol. 30, pp. 4878–4887
Summary: Formulates risk-coverage trade-offs for confidence-based rejection and abstention baselines.

[5] Guo et al. (2017): **"On Calibration of Modern Neural Networks"** **[Top ML Venue (PMLR)*]** Verified IEEE alternative: K. Fernando & C. Tsokos, IEEE TNNLS, 2022
International Conference on Machine Learning (ICML 2017), PMLR vol. 70, pp. 1321–1330
Summary: Demonstrates model overconfidence under shift; proves why confidence alone fails as a review triage.

Literature Survey: Research Papers (Part 2: Papers [6]–[10])

[6] Kim, Dan, Zhu (2024): **"Human-in-the-Loop Cyber Intrusion Detection Using Active Learning"** **[IEEE & Scopus Indexed]** DOI: 10.1109/TIFS.2024.3434647
IEEE Transactions on Information Forensics and Security, vol. 19, pp. 8658–8672
Summary: Active learning for NIDS with human-in-the-loop; directly motivates review allocation in network security.

[7] Liu, Gallego, Barbieri (2022): **"Incorporating Uncertainty in Learning to Defer for Safe Computer-Aided Diagnosis"** **[Scopus Indexed Q1]** DOI: 10.1038/s41598-022-05725-7
Scientific Reports, vol. 12, no. 1, Art. no. 1762, pp. 1–11
Summary: Incorporates uncertainty into deferral decisions, inspiring our multi-signal risk formulation.

[8] Yang et al. (2024): **"Generalized Out-of-Distribution Detection: A Survey"** **[Scopus Indexed Q1]** DOI: 10.1007/s11263-024-02117-4
International Journal of Computer Vision, vol. 132, no. 12, pp. 5635–5662
Summary: Comprehensive taxonomy of OOD detection techniques across feature and representation spaces.

[9] Koh et al. (2021): **"WILDS: A Benchmark of In-the-Wild Distribution Shifts"** **[Top ML Venue (PMLR)*]** Verified IEEE alternative: L. Ruff et al., Proc. IEEE, 2021
International Conference on Machine Learning (ICML 2021), PMLR vol. 139, pp. 5637–5664
Summary: Evaluates model degradation across real-world shifts, motivating our chronological split design.

[10] He, Kim, Asghar (2023): **"Adversarial Machine Learning for Network Intrusion Detection Systems: A Comprehensive Survey"** **[IEEE & Scopus Indexed]** DOI: 10.1109/COMST.2022.3233793
IEEE Communications Surveys & Tutorials, vol. 25, no. 1, pp. 538–566
Summary: Comprehensive survey on ML vulnerabilities in intrusion detection; provides cybersecurity context.

Comparative Study: Existing Paradigms vs Proposed Framework

Evaluation Dimension	Random Audit	Pure Confidence [4]	OOD Filter [8]	Learn-to-Defer [2]	Proposed Framework
Prediction Uncertainty	No	Yes (Sole Signal)	No	Optional	Yes (Normalized Probability)
Out-of-Distribution (OOD)	No	No	Yes (Sole Signal)	Rarely	Yes (Mahalanobis Distance)
Distribution Drift (PSI)	No	No	No	No	Yes (Feature-Level PSI)
Historical Error Proximity	No	No	No	Implicit	Yes (Decile Mistake Bins)
Explicit Capacity Budgets	Implicit	Manual Tuning	Hard Cutoff	Theoretical	Yes (1%, 2%, 5%, 10%)
Temporal Shift Evaluation	N/A	Rarely	Partial	Rarely	Yes (Chronological Split)
Adaptive Weight Scaling	No	No	No	Static Cost	Yes (Drift & Feedback Adaptive)

Key Distinction: Prior paradigms isolate confidence or OOD detection without explicit capacity slicing. Our proposed framework unifies four complementary signals into a single ranking mechanism evaluated directly under chronological distribution shift.

Users of the System & Primary Stakeholders

ML Engineer / Practitioner

• Configures and trains baseline models (XGBoost, Random Forest, Logistic Regression) on historical network flows.

• Fits training-time preprocessing pipelines and establishes reference centroids and covariance matrices for OOD detection.

• Monitors feature-level drift statistics (PSI) and configures initial heuristic risk signal weights.

• Inspects validation metrics (ROC-AUC 0.9228) and evaluates model overconfidence characteristics.

Data Reviewer / SOC Analyst

• Receives prioritized, high-risk network flow predictions routed to the human-review queue within operational budget (1%–10%).

• Inspects flagged instances displaying severe confidence degradation, feature drift, or anomalous packet patterns.

• Provides ground-truth audit verdicts on intercepted borderline and novel attack instances.

• Benefits from focused triage that mitigates alert fatigue and prevents unreviewed high-confidence ML misclassifications.

Evaluator / Research Auditor

• Executes systematic comparative evaluations across review allocation strategies (Random, Confidence, OOD, Static, Adaptive).

• Benchmarks error capture rate and review efficiency (errors intercepted per 1,000 human audits).

• Audits system robustness under chronological distribution shift against simulated oracles.

• Verifies compliance with data integrity constraints (SHA-256) and audit logging standards.

Functional Requirements: Data, Models & Signals (FR1–FR7)

FR1: Raw Ingestion & Schema Verification [100% Implemented]
Verify 500K sample integrity against 14GB source via SHA-256 fingerprints. | **Mapped to: src/01_test_dataset.py, src/26_verify_bigflow_sample.py**

FR2: Chronological Splitting [100% Implemented]
Partition flows temporally into 70% Train (350K), 15% Val (75K), 15% Test (75K). | **Mapped to: src/02_create_splits.py**

FR3: Data Preprocessing [100% Implemented]
Impute missing values with training medians, apply StandardScaler strictly on LR. | **Mapped to: src/03_preprocess.py**

FR4: Model Training & Validation [100% Implemented]
Train Logistic Regression, Random Forest, and XGBoost (val ROC-AUC 0.9228). | **Mapped to: src/04_train_models.py**

FR5: Prediction Signal Extraction [100% Implemented]
Extract prediction probability, confidence, and Shannon prediction entropy. | **Mapped to: src/05_generate_prediction_signals.py, src/16_shifted_test_predictions.py**

FR6: Confidence-Based Review Strategy [100% Implemented]
Rank predictions by ascending confidence across 1%, 2%, 5%, 10% budgets. | **Mapped to: src/06_confidence_review_strategy.py**

FR7: Mahalanobis OOD Distance Detection [100% Implemented]
Compute regularized Mahalanobis distance relative to 50K training flows. | **Mapped to: src/07_ood_detection.py, src/08_ood_review_strategy.py**

Functional Requirements: Triage, Scoring & Evaluation (FR8–FR14)

FR8: Static Multi-Factor Synthesis [100% Implemented]
Combine 50% confidence risk and 50% OOD percentile into baseline multi-factor score. | **Mapped to: src/09_static_multifactor_strategy.py**

FR9: Feature Drift Measurement [100% Implemented]
Calculate Population Stability Index (PSI) with 10 quantile bins across 49 features. | **Mapped to: src/10_measure_drift.py, src/11_create_drift_risk.py**

FR10: Adaptive Regime Risk Scoring [100% Implemented]
Classify drift severity into Low/Mod/High regimes and assign regime-specific weights. | **Mapped to: src/12_adaptive_risk_score.py, src/13_adaptive_review_strategy.py**

FR11: Historical Error Proximity [100% Implemented]
Bin 10 network flow features into deciles to map mistake patterns and mistake proximity. | **Mapped to: src/14_historical_error_risk.py, src/15_four_factor_risk.py**

FR12: Shifted-Test Evaluation Pipeline [100% Implemented]
Benchmark 5 strategies on 75,000 shifted flows across 4 operational budgets. | **Mapped to: src/18_shifted_review_comparison.py, src/21_review_efficiency.py**

FR13: Feedback-Adaptive Allocation Loop [100% Implemented]
Scale historical error weight based on reviewed error concentration feedback. | **Mapped to: src/23_review_feedback_loop.py, src/24_shifted_feedback_evaluation.py**

FR14: Interactive Web Dashboard Interface [Planned (0%)]
Provide interactive parameter tuning, dynamic budget sliders, and graphical metric cards. | **Mapped to: Planned for Phase-2**

Non-Functional Requirements Specification (Measured)

NFR1: Strict Reproducibility

• **Requirement:** Deterministic execution across all data splits, feature transformations, and model evaluations.

• **Measured Implementation:** Fixed random seed (random_state=42) across all scripts. Deterministic chronological sort ensures identical 350K/75K/75K splits on every run.

NFR2: Computation Time & Latency

• **Requirement:** Efficient execution suitable for high-throughput batch processing without excessive compute overhead.

• **Measured Implementation:** Full strategy comparison (src/25_final_strategy_comparison.py) executes in 1.60 seconds. End-to-end signal extraction across 75,000 instances completes in ~48 seconds on standard CPU.

NFR3: Data Integrity & Traceability

• **Requirement:** Cryptographic verification of data provenance and sample fidelity.

• **Measured Implementation:** SHA-256 fingerprint verification (src/26_verify_bigflow_sample.py) confirms that 100% of rows in BigFlow_sample.csv match the raw 14GB source dataset.

NFR4: Modularity & Interpretability

• **Requirement:** Decoupled software architecture with transparent, defensible scoring logic.

• **Measured Implementation:** 26 independent Python scripts communicate via standard CSV contracts. Risk scores are transparent weighted linear combinations of 4 explicit reliability signals.

Tools & Technologies Used

CORE LANGUAGE & RUNTIME

• **Python 3.10+:** Primary execution runtime for data processing, scientific modeling, and evaluation pipelines.

• **pathlib & os:** Robust cross-platform filesystem path resolution across local and cloud environments.

DATA ENGINEERING & STORAGE

• **pandas (v2.x):** High-performance tabular data manipulation, chronological sorting, and CSV persistence.

• **NumPy (v1.26+):** Vectorized array computations, numerical stabilizer arithmetic, and quantile calculations.

MACHINE LEARNING & MODELING

• **XGBoost (v2.0+):** Primary gradient-boosted decision tree classifier (150 estimators, max depth 6).

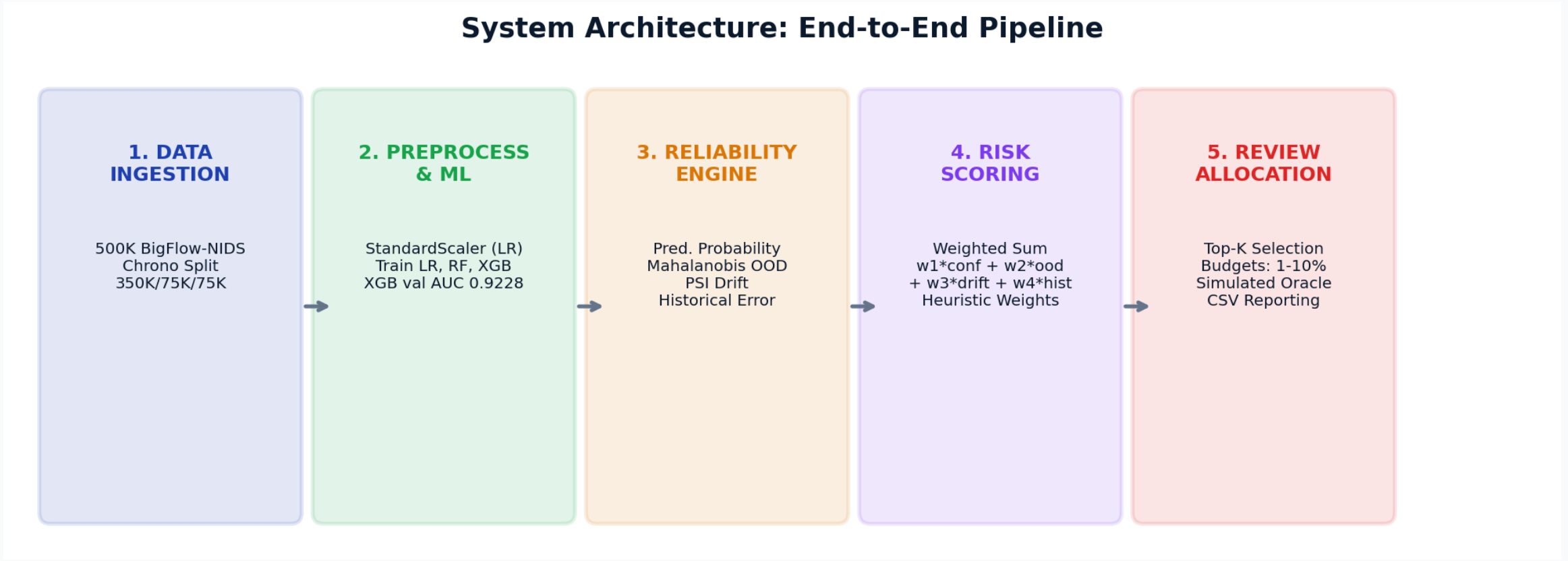
• **scikit-learn (v1.3+):** StandardScaler preprocessing, Logistic Regression, Random Forest, and ROC-AUC evaluation.

SCIENTIFIC COMPUTING & SECURITY

• **SciPy (v1.11+):** Regularized covariance matrix inversion, Mahalanobis distance, and Shannon entropy.

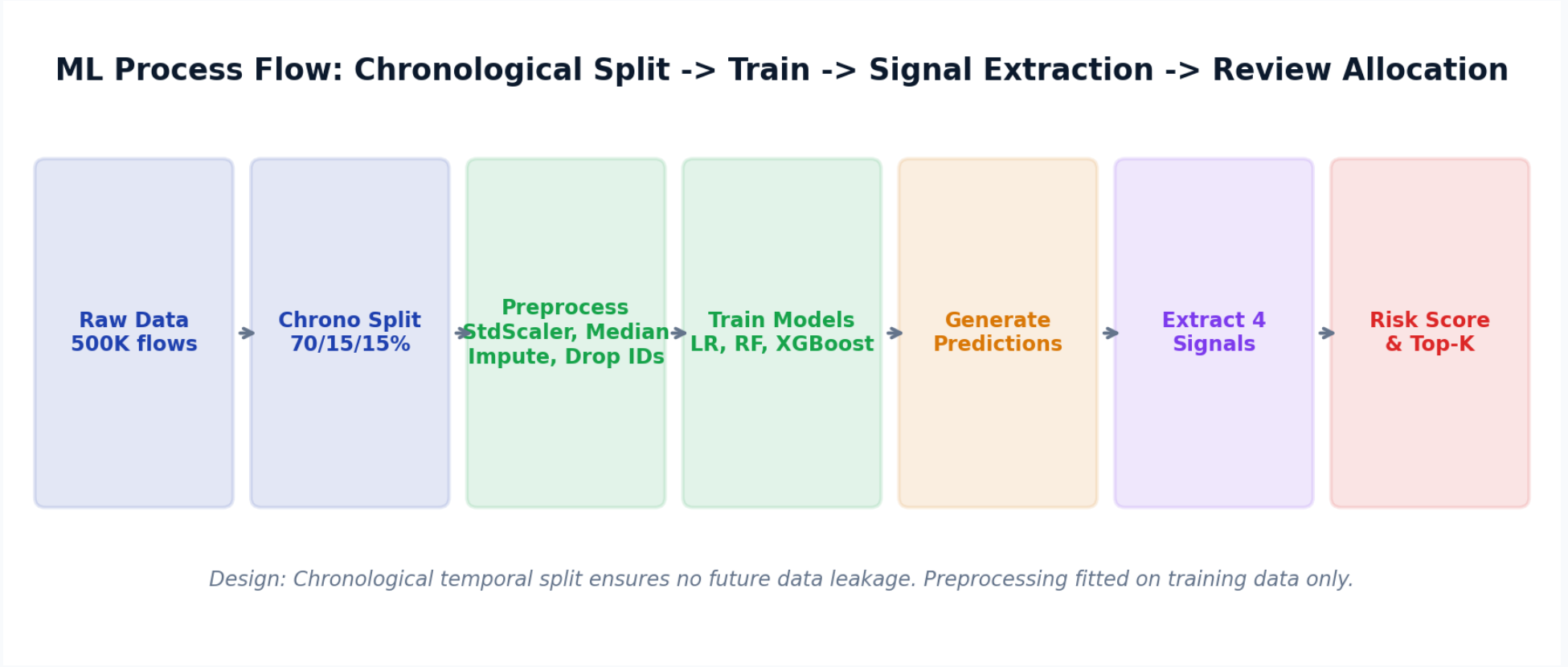
• **joblib & hashlib:** Serialized model artifact persistence (.pkl) and SHA-256 cryptographic sample verification.

System Design: Architecture Diagram



Architectural Dataflow Summary:
Data flows through five discrete stages: (1) Ingestion & temporal partitioning -> (2) Preprocessing & baseline inference -> (3) 4-signal reliability feature extraction -> (4) Composite risk scoring & priority queue creation -> (5) Review allocation to simulated human oracle within budget (1% to 10%), dispatching remainder to automated model decision.

System Design: ML Process Flow & Methodology

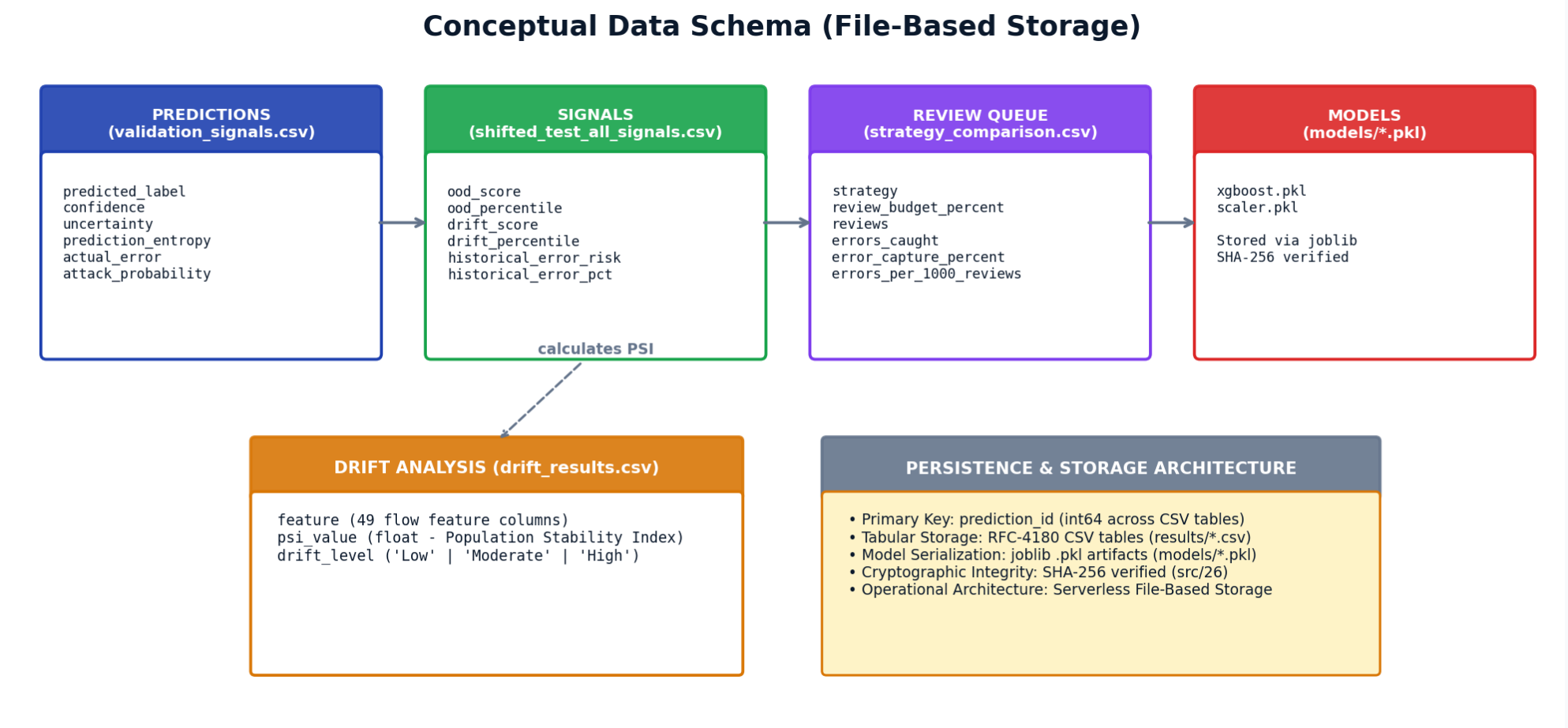


Design: Chronological temporal split ensures no future data leakage. Preprocessing fitted on training data only.

Methodological Safeguards:

Strict chronological splitting prevents data leakage. Preprocessing parameters (median imputation, feature scalers) are fitted strictly on the 350K training split. All OOD covariance calculations and historical mistake bins are derived without access to shifted test labels, ensuring genuine temporal evaluation.

Database Design: Conceptual Schema & Entity-Relationship



File-Based Data Architecture: Stored across structured CSV tables and serialized joblib artifacts. Primary keys (prediction_id) maintain relational integrity across prediction signals, drift metrics, and final review queue allocations without requiring a database server.

Benchmark Dataset & Temporal Distribution Shift

BIGFLOW-NIDS BENCHMARK (2026) [11]

• **Derived Benchmark:**
Harmonizes 4 major public NetFlow testbeds (NF-UNSW-NB15, NF-ToN-IoT, NF-BoT-IoT, NF-CSE-CIC-IDS2018) via Spark.

• **Full Dataset Scale:**
66,935,021 total network flows, 55 attributes, 32 distinct attack categories across four research testbeds.

• **Experimental Sample:**
500,000 chronological flows extracted for rigorous laptop-scale research experimentation.

• **Feature Formulation (X):**
49 numerical flow attributes including packet counts, byte volumes, duration, IAT, and TCP flags.

• **Target Formulation (y):**
Binary classification: 0 = Benign, 1 = Attack. Detailed Attack category excluded to prevent target leakage.

• **Scientific Fact:**
An offline research benchmark, NOT live production or live Internet traffic.

CHRONOLOGICAL SPLIT & SHIFT DYNAMICS

• **Training Split (70%):**
350,000 flows. Baseline attack prevalence: 6.10% (21,344 attacks). Used to fit scalers, reference distributions, and models.

• **Validation Split (15%):**
75,000 flows. Attack prevalence: 0.33% (248 attacks). Model achieves 98.996% accuracy, making exactly 753 errors (1.004% error rate).

• **Shifted Test Split (15%):**
75,000 flows from a later temporal period. Attack prevalence surges to 6.30% (4,724 attacks). Model error rate surges to 4.569% (3,427 errors).

• **Observed Distribution Shift:**
Attack rate increases by 19.05x; error count increases by 4.55x (from 753 to 3,427 errors). Yet average confidence drops only from 98.65% to 96.82%, demonstrating severe model overconfidence.

Model Training & Validation Performance (Verified)

Model Architecture	Accuracy	Precision	Recall	F1-Score	Validation ROC-AUC
Logistic Regression (with StandardScaler)	0.9606	0.0376	0.4435	0.0693	0.7499
Random Forest (150 trees, max depth 20)	0.9801	0.0708	0.4153	0.1210	0.8526
XGBoost Classifier (Primary Selected)	0.9900	0.0401	0.0887	0.0552	0.9228

VALIDATION PREDICTION METRICS & SELECTION RATIONALE

• **Primary Model Selection:** XGBoost was selected as the operational base classifier due to its superior discriminative power, achieving a validation ROC-AUC of 0.9228 (compared to 0.8526 for Random Forest and 0.7499 for Logistic Regression).

• **Exact Validation Errors Verified:** Across all 75,000 validation instances, XGBoost makes exactly 753 prediction errors (527 False Positives, 226 False Negatives), establishing a baseline error rate of 1.004%.

• **Class Imbalance Reality:** Due to severe class skew (0.33% attack prevalence), raw precision and F1 appear low across all models; ROC-AUC is the true ranking metric.

• **Baseline Confidence Stability:** On validation data, average XGBoost prediction confidence is 0.9865, reflecting high certainty in the unshifted operating regime.

Implementation Demo: Module Status (92.9% Verified)

Implementation Module Status (26 Scripts)

Module	Scripts	Status	Complete
Data Ingestion & Verification	01, 26	100%	
Chronological Splitting	02	100%	
Preprocessing (clean, impute, scale)	03	100%	
Model Training (LR, RF, XGBoost)	04	100%	
Prediction Signal Generation	05, 16	100%	
Confidence-Based Review	06	100%	
OOD Detection (Mahalanobis)	07, 08	100%	
Static Multi-Factor Strategy	09, 15	100%	
Drift Measurement (PSI)	10, 11	100%	
Adaptive Risk Scoring	12, 13	100%	
Historical Error Proximity	14	100%	
Shifted-Test Evaluation	16-22	100%	
Feedback-Adaptive Strategy	23, 24, 25	100%	
Interactive Dashboard / UI	—	0%	

13 of 14 modules fully implemented (93%). Interactive UI: Phase-2.

Implementation Audit Result: 13 of 14 Modules Implemented (92.9% Complete)

The implementation consists of 26 sequentially executable Python scripts covering end-to-end data ingestion, model training, 4-factor signal extraction, and review allocation. Only FR14 (interactive web dashboard) is reserved for Phase 2.

Implementation Demo: Signal Extraction & Shift Evidence

4 EXTRACTED RELIABILITY SIGNALS

• **1. Prediction Uncertainty (s_conf):**
Derived directly from XGBoost predict_proba output as (1 - max(P(y|x))). Ranks predictions near the decision boundary.

• **2. Mahalanobis OOD Distance (s_ood):**
Computed using regularized pseudo-inverse covariance (eps=1e-6) relative to 50K training flows, converted to percentile ranks.

• **3. Feature Drift Score (s_drift):**
Identifies features with PSI >= 0.10. Scores instances by absolute z-score deviation weighted by feature PSI.

• **4. Historical Error Proximity (s_hist):**
Projects validation instances into 10-decile error risk bins across 10 network flow attributes based on training mistake patterns.

MEASURED DISTRIBUTION SHIFT EVIDENCE

• **Error Surge (4.55x):**
Validation errors: exactly 753 (1.004%). Shifted test errors: exactly 3,427 (4.569%). Error count increases by exactly 4.551x.

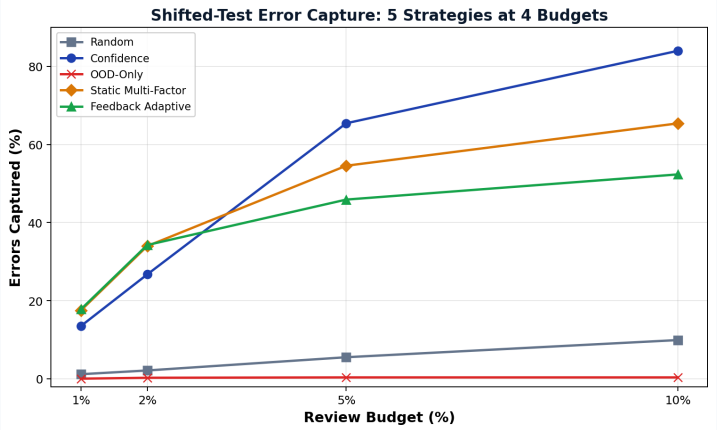
• **Attack Prevalence Surge (19.05x):**
True attacks increase from 248 (0.33%) in validation to 4,724 (6.30%) in the shifted test period.

• **Feature Drift (PSI Measurement):**
Population Stability Index reveals severe drift across key flow metrics: SHORTEST_FLOW_PKT (PSI=0.766), MAX_IP_PKT_LEN (PSI=0.757), and IN_BYTES (PSI=0.604).

• **Model Overconfidence:**
Average model confidence drops by only 1.83% (from 98.65% to 96.82%), confirming that XGBoost fails silently on novel, drifted attack patterns.

Empirical Results: Shifted-Test Strategy Benchmark

Budget	Random Audit	Pure Confidence	OOD-Only Filter	Static Multi-Factor	Feedback Adaptive
1.0% (750)	**1.17%** 40 errs | 53.3/k	**13.51%** 463 errs | 617.3/k	**0.03%** 1 err | 1.3/k	**17.51%** 600 errs | 800.0/k	**17.80%** 610 errs | 813.3/k
2.0% (1,500)	**2.13%** 73 errs | 48.7/k	**26.73%** 916 errs | 610.7/k	**0.26%** 9 errs | 6.0/k	**33.99%** 1,165 errs | 776.7/k	**34.26%** 1,174 errs | 782.7/k
5.0% (3,750)	**5.52%** 189 errs | 50.4/k	**65.45%** 2,243 errs | 598.1/k	**0.35%** 12 errs | 3.2/k	**54.57%** 1,870 errs | 498.7/k	**45.90%** 1,573 errs | 419.5/k
10.0% (7,500)	**9.92%** 340 errs | 45.3/k	**84.01%** 2,879 errs | 383.9/k	**0.35%** 12 errs | 1.6/k	**65.42%** 2,242 errs | 298.9/k	**52.35%** 1,794 errs | 239.2/k



CRITICAL SCIENTIFIC INSIGHTS & HONEST INTERPRETATION

• **Tight Budget Advantage (1%–2%):** Feedback Adaptive and Static Multi-Factor capture more errors than confidence alone (17.80% vs 13.51% at 1%; 34.26% vs 26.73% at 2%). Multi-factor triage is most beneficial under tight human review constraints.

• **Broad Budget Crossover (5%–10%):** At higher review budgets, pure Confidence ranking catches up and captures more total errors (65.45% at 5%, 84.01% at 10%). Confidence targets broad decision-boundary ambiguity across the general distribution.

• **OOD Standalone Failure (Honest Negative Result):** Mahalanobis OOD distance alone underperformed random audit across all budgets (0.03% vs 1.17% at 1%), demonstrating that feature novelty in network flows does not correlate with classification mistakes.

• **Core Defensible Finding:** Strategy effectiveness is budget-dependent. No single strategy dominates across all budgets, establishing that human-review allocation mechanisms must be selected to match operational review capacity.

Limitations & Phase-2 Research Roadmap

IDENTIFIED RESEARCH LIMITATIONS

• **Simulated Review Oracle:**
Review evaluation uses test ground-truth labels rather than human subjects; real-world reviewer fatigue, latency, and human error are unmodeled.

• **Heuristic Signal Weighting:**
Weights are experimentally defined rather than formally learned through bilevel optimization or reinforcement learning.

• **Single-Dataset Benchmark:**
Current findings are evaluated exclusively on BigFlow-NIDS; cross-domain validity on healthcare and financial tabular data remains to be tested.

• **Historical Error Leakage Consideration:**
Historical error proximity currently uses training-set mistakes and would benefit from out-of-fold cross-validation estimates.

• **No Universal Dominance:**
Multi-factor allocation does not dominate confidence at large budgets (5%–10%).

PHASE-2 RESEARCH ROADMAP

• **Dataset-Agnostic Ingestion:**
Extend framework to automatically profile and process arbitrary tabular datasets across credit risk, fraud, and medical diagnosis.

• **Post-Hoc Probability Calibration:**
Integrate temperature scaling and conformal prediction [4], [5] to provide distribution-free confidence guarantees.

• **Instance-Level Explainability (SHAP):**
Generate feature attribution explanations for each routed prediction so human reviewers understand why an alert was escalated.

• **Cost-Sensitive Optimization:**
Replace heuristic linear weights with learned optimal routing policies incorporating asymmetric financial penalty matrices.

• **Interactive Demonstration UI (FR14):**
Develop a lightweight web interface for live budget adjustments and review queue visualization.

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